



Design and Mixed-Reality Teleoperation of a Quadruped-Manipulator Robot for SAR Tasks

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Abstract. Technological development within the robotics field has made it possible to generate great advances, developing complex bio-inspired systems, such as robots with legs. The same ones have great applicability within search and rescue (SAR) tasks. This type of robot stands out for its extraordinary ability to move within unstructured environments, overcome obstacles and adapt to different terrains. However, quadruped robots have gained space in the SAR-Tasks field over the last few years. Most implementations are limited to collecting information using different sensors, such as cameras, lasers, or microphones. This article seeks to simulate, implement and carry out the teleoperation through Mixed-Reality (M-R) of a quadruped robot equipped with a manipulator with six degrees of freedom, for which the ARTU-R robot (A1 Rescue Tasks Unitree Robot) has been used. One of the main contributions of this work focuses on improving the efficiency in executing tasks of handling and transporting medical equipment in post-disaster situations, using Mixed Reality with the ARTU-R robot and a robotic manipulator. To develop this proof of concept, Matlab has been used as a computational tool for optimizing the workspace of the integrated robot. Simulations have been carried out on Gazebo in reconstructed post-disaster environments to validate the robot's functionality. At the same time, the effectiveness of the M-R system has been verified with field tests executing medical assistance tasks with the robot. The main results show a 21% increase in the efficiency of performing complex handling tasks using the proposed M-R system compared to conventional interfaces and the efficiency of using quadruped robots with manipulators for medical assistance tasks.

Keywords: Quadruped robot · Mixed reality teleoperation · ROS · Arm manipulator · Walking robots · Search and rescue · Gazebo

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1 Introduction

The study of quadrupeds robots focused on search and rescue (SAR) tasks has evolved to the multiple advantages provided over conventional mobile robots (robots with caterpillar tracks, wheels) [1]. Mainly their ability to move through unstructured terrain stands out, with debris [2,3].

The importance of quadruped robots as intervention platforms in post-disaster environments has been increasing in recent years, carrying out highly relevant developments in search and rescue. An example is the one developed by the authors, which focuses on detecting buried victims with thermal images using the Unitree A1 platform as a means of displacement in the environment [4].

In this preliminary study phase, the advantages of using a quadruped robot in this environment have been analyzed. However, it has been limited to data capture, as in most developments in this field [5–10]. Therefore, in this new phase of development, integrating a manipulator with six degrees of freedom has been proposed to carry out complementary tasks such as assistance to victims with medical equipment [11]. For this implementation, a workspace optimization phase has been previously carried out using Matlab and Gazebo simulations to verify the prototype's functionality.

The control of these systems is complex to conventional interfaces, such as screens with a mouse and keyboard, mainly due to the remote operator's lack of perception of the environment or the simultaneous handling of robot variables. Some developments have sought to address this problem by providing teleoperation solutions. However, they are quite rudimentary and invasive for the operator, like the developments carried out in [12,13].

The main contribution of this article focuses on the implementation of a Teleoperation method of this newly implemented platform through a Reality-Rixta (MR) system where the operator becomes aware of the environment through a virtual model projected through Hololens glasses, being able to carry out controlled movements in the workspace of the articulating robot, through a visual mark on the end of the robot.

The system has been shown to improve the efficiency in the execution of first aid equipment transport tasks due to the contribution of Mixed Reality since the operator can identify the type of object and perhaps its possible fragility, deformability, etc. and carry out the task with greater confidence.

The validation tests have been carried out in post-disaster scenarios reconstructed according to the specifications of the NIST (National Institute of Standards and Technology).

This paper is structured as follow. Section 2 describes the most relevant works in state-of-the-art; Sect. 3 shows materials and methods in detail. Section 4 containing the experiments and results. Finally, Sect. 5 presents the main findings.

2 Related Works

Figure 1 shows the articles increase, with topics related to Quadruped Robots for Search and Rescue; in the web of knowledge, it shows an mean increase of around 13% of articles per year until 2018. However, in the last three years, the number of articles published has increased significantly, up to an incremental rate of up to 100 compared to 30 in previous years. This indicates the importance of this type of robot within search and rescue. In [14] reference is made to the growth of these quadrupeds, while in studies such as [15], a follow-up and justification of the evolution of this type of mechanisms is made up to the present day.

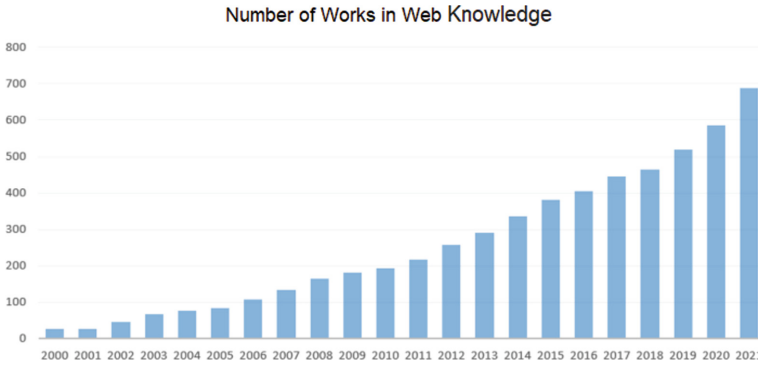


Fig. 1. Numbers of works in the web of knowledge including quadruped robots for search and rescue as topic.

Within the first phases of the study and analysis of quadrupedal robots, explorations and balance studies have been carried out [16], the behaviour of gait modes, trajectory planning, and commercial platforms such as ANYmal [17].

The variability of scenarios and complex tasks demand the remarkable capacity of robotic systems. It is convenient to operate with suitable actuators; depending on the field and the execution [18], there are differences between electrical, hydraulic, and pneumatic actuators. Regarding handling, in [19], it is about exposing the added difficulty that deformable objects have, a subject to study due to the uncertainty in SAR tasks and the variety of characteristics that can be found.

The teleoperation of the set is added as a complement to the control, also studied in this article, with references to previous works such as [20], where the problem of communication delays is also shown.

Implementations have been carried out with quadrupedal robots for search and rescue tasks focused on different objectives such as facilitating objects to the required place, picking up objects and taking them to a safe place or just obtaining data.

The environments that the robot faces can be very hostile. In addition to the topology of the terrain, they also affect its characteristics. Another studies works on an adaptive control to all types of surfaces, such as snow, water or leafy vegetation, making it difficult to perceive the scene [21].

Work times in the field can be very long, so good management and optimization of the energy consumed [22] is key to tackling SAR tasks. In case of having to cover much ground or collect very different types of data, it is interesting to study the use of fleets of robots. Studies have been carried out in terms of planning, assistance and communication with the human in these cases in which several quadrupeds are worked on simultaneously [23].

Among the most recent developments in this field are ALMA and RoLoMa, which use the ANYmal quadruped robot as a base, equipped with a 6 DoF manipulator (Jaco²), on which different analyzes have been carried out, mainly dynamic and stability, as well as door opening tasks [24,25]. For all types of loads, it is necessary to have control over the influence of the forces generated by the arm when transporting them and also when carrying out the pick and place task, as studied in the article [26]. While other centaur-type robotic systems have focused on locomotion adaptation with heavy load [27]. For this type of task, implementations of hydraulic arms have been studied in conjunction with also hydraulic quadrupeds [28].

Due to the innumerable scenarios in which to carry out tasks, we have to minimize the uncertainty and errors that are made due to ignorance of the object we want to manipulate. Some of the main methods to control these complex systems use deep learning, as in the development carried out in [2], which uses the ALMA robot based on the ANYmal business model. Studies such as the one by P. Mitrano in the journal “ScienceRobotics” [29] deal with the problem of operating with deformable objects or with unexpected characteristics in order to know if the model implemented in the robot is valid.

There are many works on the control and modelling of manipulative quadrupeds, but the way of interacting and obtaining feedback is in a less developed phase. Remote control or touch control [30] are the most common. The main drawback in the development of teleoperation is the delays that may occur in the set of peripherals since they affect the model and it ceases to be robust. There is some study that integrates inertial sensors and virtual reality gloves [20], but it is the articles and field of study where, for the moment, less has been investigated.

3 Materials and Methods

3.1 Materials

The experimental development has been carried out in the facilities of the Automatic and Robotics Center of the UPM. The Robotics platform used for this development is a quadruped robot of the Unitree trademark, equipped with a sensory system and a robotic arm described in Table 1.

Table 1. Elements and components of the quadruped robot and M-R system.

Component	Amount	Description
Hololens (1st generation)	1	Mixed-reality glasses
Unitree A1	1	Quadruped robot
RP-Lidar	1	Lidar sensor
WX250S	1	6DoF robot arm
Nvidia Jetson Xavier-Nx	1	On-board embedded system
Real-sense	1	RGB-Deph sensor

3.2 Robot Design and Modeling

The modelling and design of the set have been developed in Matlab. The direct-kinematic models have been obtained independently for the manipulator and one leg of the quadruped. In both cases, the Denavit-Hartenberg parameters are calculated; they are shown in the Tables 2 and 3 for the convenient reference system. The correct location of each leg of the model has been carried out using translations and rotations over Table 2.

The local reference systems of each leg and the robotic arm is located in the robot torso, as shown in Fig. 2(a). The entire assembly reference system has been placed in the centre of the A1 body, below the arm system.

Table 2. One leg Denavit-Hartenberg's parameters. 3DOF

Thetha (deg)	d (mm)	a (mm)	Alpha (deg)
q1 + 90	55	0	-90
q2	55	200	0
q3	0	215	0

Table 3. 6DOF manipulator's parameters

Thetha (rad)	d (mm)	a (mm)	Alpha (rad)
q1	104.43	0	$\pi/2$
q2 + $\pi/2 - 0.2$	0	254.95	0
q3 + 0.2	0	0	$\pi/2$
q4	250	0	$-\pi/2$
q5	0	0	$\pi/2$
q6	131	0	0

Both the robotic arm and the height and position of the quadruped are predominant in determining the optimal workspace. This is created from iterating

and saving the values (x, y, z) obtained from slight variations in the 6 degrees of freedom that the arm has.

Kinematic-ally there is access to a spherical but flattened shape, which covers from a radius of 650 mm to 740 mm. It is shown in red colour in Figs. 2(a)–(b). According to the manufacturer's recommendation regarding the optimal work zone, it is recommended to work at 70% of the maximum range; this is shown in the green zone.

There is a high risk of collision with the robot's head or rear. The different colours of the points in Fig. 2 were established to differentiate the manipulation zones that the robot can reach. Therefore, the green area is optimal for carrying out manipulation tasks. However, there are areas where it has not been considered for manipulation due to the manufacturer's recommendation or the generation of high torques. This is the case in the red zone.

Finally, and for stability reasons, it has been considered that the manipulator has more performance in the lateral areas than in the front and rear. An ellipsoid with its semi-major axis in the transverse direction of A1 is shown in blue.

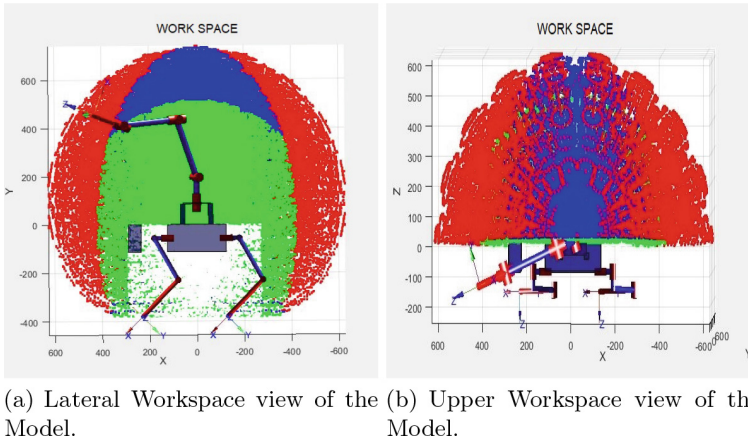


Fig. 2. Kinematic model and workspace of WX250S manipulator integrated to Unitree A1.

3.3 ROS-Gazebo Implementation and Simulation

Several test environments have been developed in ROS-Gazebo to test the movement of the entire kinematic assembly and the manipulator movement planning systems. Figures 3(a)–(c) show the simulated environments recreated based on post-disaster scenarios. These environments include small debris, unstructured soil, victims, and the quadruped robot model and the robotic arm integrated through an assembly carried out in *.xacro* files. Figures 3(b)–(d) correspond to the perception-planning systems of the respective Gazebo environments, where

the robot model is shown in the current position (in black), the target position of the arm (in Orange).

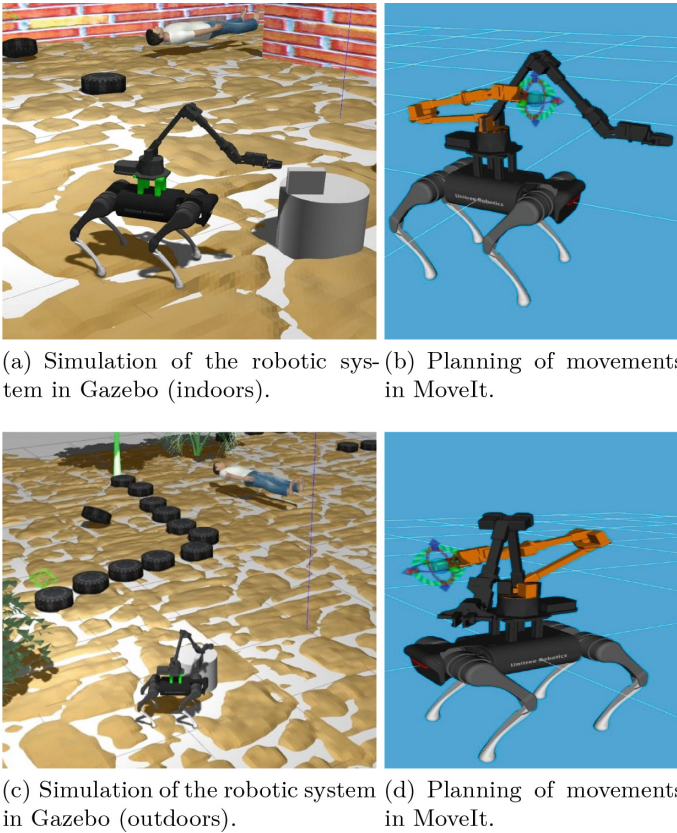


Fig. 3. Simulation of the quadruped robot together with the manipulator and Movement Planning (MoveIt) in different Gazebo environments for the execution of manipulation tests.

3.4 Mixed-Reality Teleoperation

Figure 4 shows the operating scheme of the teleoperation system. The operator uses Hololens glasses as a means of human-robot-environment interaction. In the glasses, both the physical environment and the hologram model of the robot are observed. It has a blue sphere at its end, which the user can manipulate through gestures to move the arm towards an area of interest. The joint positions of the simulated robot are sent to the controller of the real robot to execute the movement.

The movements of the fingers have been detected due to the functions of Holo-Toolkit, precisely the Manipulation Mode function. Which captures the union of the fingers showing the interaction with the virtual objects (end of the robot), then the inverse kinematics is calculated, and the positions are sent to the real robot via Rosbridge.

The mixed reality interface also has buttons that allow the gripper to open-close and execute movements to predefined positions (home and sleep) and a small screen where the image of the robot is transmitted outdoors. All these buttons can be manipulated by the joint gesture of the fingers (thumb-index) of the operator wearing the glasses.

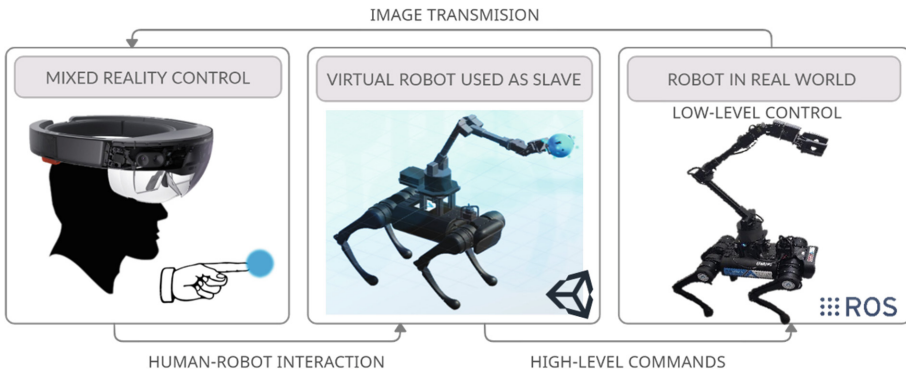


Fig. 4. Systematic scheme of teleoperation of the robot-manipulator through mixed reality.

The strategy contemplated for transporting first aid equipment using the implemented robot and the M-R system is based on an approach phase, where the robot with the arm in the collected position without load moves, minimizing its gravity centre towards point A. As long as it is not being manipulated, the ideal is to pick up the manipulator and work with a control model that dampens the oscillations generated in the quadruped body since it is the area where it joins the arm.

Once in the area, the user moves the arm remotely (at 5 m maximum distance) until taking the object with the gripper. The lateral posture is the most favourable to proceeding to load an object since the quadruped can adopt configurations with more stable joints to support the object's weight. This is due to the approach distances necessary to pick-place since they are longer longitudinally and generate more torque.

Once the object has been picked up, the arm acquires the transport position, reducing the gravity centre until the robot moves towards the victim's area (point B). Finally, there is teleoperated to deliver the equipment.

4 Experiments and Results

The development of tests to validate the functionality of the joint robotic system implemented (Quadruped Robot + Robotic Arm) and the functionality of the teleoperation system. The main tests to analyze the functionality of the robotic system have been based on manipulating objects with different weights, taking them from point A to point B.

The teleoperation system through mixed reality has been evaluated based on the versatility to control of the robotic system through teleoperation and has been contrasted with conventional interfaces. The performance tests can be seen in more detail in Appendix A.

4.1 Evaluation of the Mixed Reality System for Teleoperation

Tests have been developed with two types of interfaces. The first is the M-R Teleoperation system, and the second is a conventional interface (RVIZ); both have the same purpose, to perform movements on the manipulator robot of the robotic system, considering the workspace and the self-collision. Testers have used both interfaces to perform the same task, which is to move the end of the robot from point A to point B, marked with an X on the ground.

Figures 5(a)–(b)–(c) show the hand of the person taking the blue ball and performing the movements in space. These Figures show the superimposition of the robots' virtual model (grey colour) on the real robot (black colour); in such a way that as the blue ball moves, the virtual model adjusts the kinematic robots' positions to send them to the real robot, carrying out the teleoperation. In this first phase of experimentation, up to 200 ms in teleoperation has been delayed.

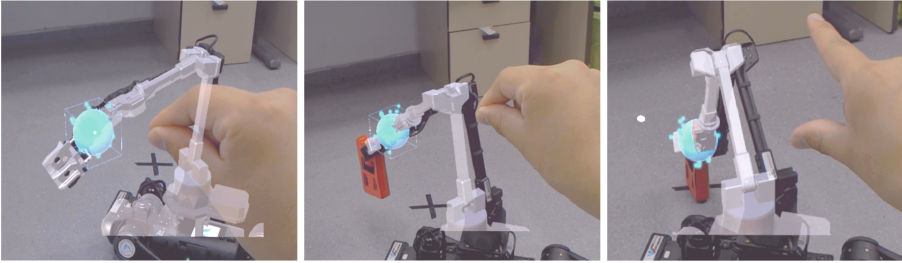
Figures 5(d)–(e) show the box diagrams of the data collected, where participants' experiences show that the implemented system generates a better experience for immersive situations by 15% and increases confidence when executing movements with the robotic arm by 23% compared to conventional interfaces.

4.2 Pick and Place Test Using M-R System

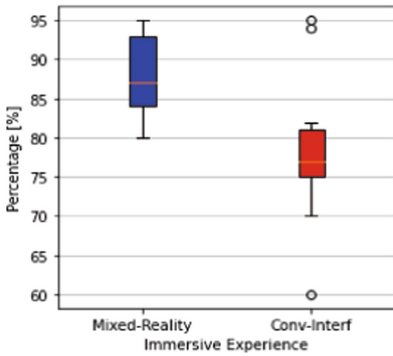
The next phase of experiments consisted of moving the robot's end from point A, taking a medical-type object and depositing it at point B, marked with an X on the ground. The transported objects are a first aid kit and a bottle of alcohol for healing.

Figure 6 [a–d] shows the different stages to transport a first aid kit, starting from $t = 0$ s. The first test object weights 100 g. The object has been reached and picked up in the first 10 s, while to reach the destination position and drop it, an additional 10 s has been required.

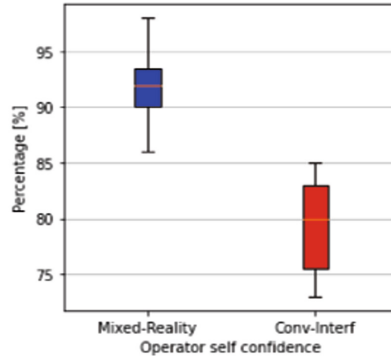
On the other hand, Fig. 6 [e–h] shows the steps to take from point A to point B, a bottle of alcohol weighing 150 g, unlike the previous test object, it is held in the first 7 s due to its cylindrical shape, but its transport to the destination point takes 12 s more because the operator requires greater precision in the movements to avoid the fall of the object with liquid that can damage the robot.



(a) Teleoperation in home (b) Teleoperation in 2nd (c) Teleoperation to final position.
position. Pose.



(d) Immersive experience analysis.



(e) Operator confidence analysis.

Fig. 5. Tests to evaluate the Mixed-Reality efficiency system, the operator holds the robots end (blue ball) and using his gestures, moves it towards the destination (X Mark).

Therefore, the greater the weight or, the more delicate it is, the operator will require more precise movements and the demand for time to carry out these tasks will increase proportionally to the weight.

4.3 Victim Assistance Test in Reconstructed Post-disaster Environments

The third phase of tests has been carried out in a test environment reconstructed according to NIST specifications, consisting of small debris and victims (mannequins and people). In this scenario, the task carries a first-aid kit from point A to point B (victim). Figures 7 [a–c] show the different stages of the trajectory executed by the robot. In the first instance, it approaches and takes the first aid kit, places the arm in the transport position and moves through the environment until reaching the victim. Finally, he extends his arm and drops the package near the victim. The entire process is teleoperated by the robot from the remote control and the arm from the M-R system.

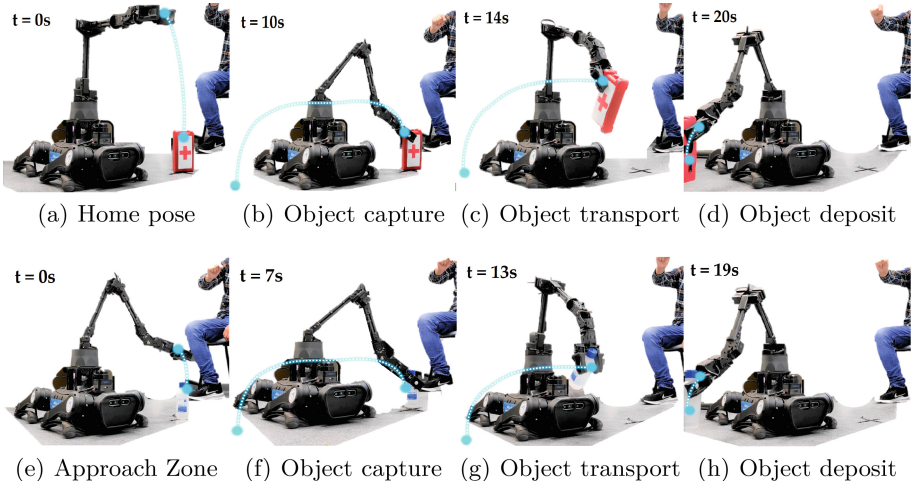


Fig. 6. Tests of transportation medical assistance objects (First kit aid [a-d] and alcohol [e-h]), through M-R Teleoperation.

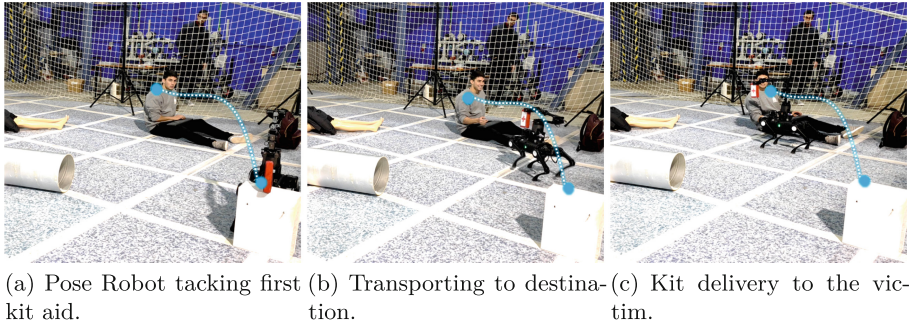


Fig. 7. Teleoperation and metrics evaluation R-M teleoperation system for robot's assistance with medical equipment for a victim.

Based on the experiences collected by the operators, the following graphs have been established (Figs. 7(d)–(e)), which describe the efficiency of executing tasks based on the operators' complexity and previous experience.

Figure 7(d) shows the efficiency evolution based on the task complexity (defines if a task is easy to perform or not) for both cases, executions with Mixed Reality and conventional interface. Both show high efficiency for simple tasks like moving the arm from one point to another. However, for more complex tasks such as object manipulation, the mixed reality system is superior by 21%.

Figure 7(e) represents the evolution of the efficiency to execute tasks based on training with both types of interfaces. Initially, the conventional interfaces show to be better since they are intuitive. However, after a series of training sessions (20 tests), both interfaces are better. However, there is a turning point where mixed reality surpasses the conventional. This is due to several factors, one of which is the immersive experience generated by using these devices.

5 Conclusions

The implementation of the quadruped-manipulator joint kinematic modelling in Matlab has allowed calculating and optimizing the work area of the manipulator robot in such a way that an optimal zone has been established based on the kinematic characteristics and approach strategies of the robot for object manipulation.

The ROS simulation environment (Gazebo) has shown great potential and utility to recreate physical conditions and integrate the joint model designed. This has made it possible to validate the functionality before physical integration and the execution of movements without manipulators collision in the workspace, using the MoveIt packages.

Mixed-Reality interfaces as a means of teleoperation concerning conventional interfaces have shown potential advantages. Mainly in the increased confidence of operators for decision making and the significant improvement in efficiency to perform complex handling tasks with objects with variable geometry and weight.

Previous experience and training of operators with mixed reality greatly influence the efficiency with which tasks are executed. Initially, the interfaces used were better for inexperienced operators. Still, as a progressive training, the evolutionary results place the M-R interfaces as the most suitable solution to teleoperate complex robotic systems in search and rescue tasks.

In future works, the line is left open to evaluating other interfaces such as virtual reality and executing synchronized movements between the quadruped robot and the arm to perform the same tasks.

Appendix A

Video of Teleoperation tests through the Mixed-Reality Execution system for the implemented robot https://www.youtube.com/watch?v=_uN22XUTWjo.

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On the Kinematic Characteristics of Walking Mechanisms of Mobile Robots with Walking Propulsion Devices

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Abstract. Various kinematic schemes of walking propulsion devices of mobile robots with different number of drives are analyzed. The errors in the implementation of the program movement of the propulsion device foot are estimated with the help transfer functions. With the use of transfer functions, matrices characterizing the specific mechanism of the mover are applied. The scalar indicator of the average accuracy of the realization of both position and speed of the foot in the program mode of movement is considered. A specific “zoomorphic” propulsion device is considered and an estimate of its accuracy is given depending on the size of the links. The obtained dependences allow the formulation of the problem of optimization of geometric parameters of mechanisms.

Keyword: Mechanism. Propulsion device. Matrix of transfer functions. Scalar criterion

1 Introduction

Mobile robots with walking propulsion devices are known, which differ from each other both in the number of control drives and in their kinematic schemes.

Single-drive walking mechanisms, as a rule, are cyclic mechanisms, which are characterized by a well-defined trajectory of the supporting foot relative to the body of the walking robot. An example of such robots is the robot “Vosminog” [1] (Fig. 1) with a four-link walking mechanism equipped with a corrector gearbox.

The principle of operation of such a propulsion device and similar ones [2] is based on the control of the rotational movement of the driving crank. Therefore, despite the well-defined trajectory of the foot, it is possible to control the law of its movement of the trajectory by programmatically changing the angular velocity of the leading link - the crank. These methods can be provided both by using additional mechanisms included in the kinematic chain of the propulsion device drive [3], and by controlling the movement of the drive motor.